

ONLINE SHOP

Group Project
Requirements & Methodology



PRESENTATION'S AGENDA

- Project Overview
- Project Scope
- Functional Requirements
- Non-Functional Requirements
- Use Cases
- Methodology
- Summary

PROJECT OVERVIEW

This project focuses on developing an **Online Shopping System** that allows users to browse, select, and purchase products through a simple and intuitive web interface.

The system provides essential features for customers (such as browsing, searching, and ordering)

administrative functions (such as managing products and viewing orders). The main objective is to build a functional prototype that demonstrates the complete workflow of an online shop.



PROJECT SCOPE

In-Scope Features

- User registration and authentication.
- Product browsing and search functionality.
- Shopping cart management.
- Order placement and checkout process.
- Administrative management of products and orders.

Out-of-Scope Features

- Integration with real payment gateways (ex: PayPal, Stripe).
- Delivery tracking with external logistics systems.
- Development of a native mobile application.

FUNCTIONAL REQUIREMENTS

ID	Requirement	Description
FR1	User Registration/Login	The system allows users to register an account and log in securely.
FR2	Product Catalog	Displays available products with details such as name, image, and price.
FR3	Add to Cart	Users can add items to their cart and view the selected products.
FR4	Checkout Process	Users can proceed to checkout and confirm their order.
FR5	Admin Management	Administrators can manage products and review customer orders.

NON-FUNCTIONAL REQUIREMENTS

ID	Requirement	Description
NFR1	Usability	The interface should be user-friendly and easy to navigate.
NFR2	Performance	The system should respond to user requests time ≤ 2 seconds.
NFR3	Security	User credentials and data must be stored securely.
NFR4	Scalability	The system architecture should allow future feature expansion.
NFR5	Compatibility	The system should be compatible with major browsers, operating systems, and devices (desktop, tablet, mobile).
NFR6	Reliability	The system should handle errors without crashing or losing data.

USE CASE

- **Register and Login**
 - **Actor:** Customer
 - **Description:** The user registers for an account or logs in to access personalized features such as cart and order history.
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- **Browse and Search Products**
 - **Actor:** Customer
 - **Description:** The user views available products and uses search or filter options to find specific items.

USE CASE

- **Add to Cart and Checkout**
 - **Actor:** Customer
 - **Description:** The user adds products to the shopping cart and proceeds to checkout to confirm the purchase.
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- **Manage Products**
 - **Actor:** Administrator
 - **Description:** The admin can add, update, or delete products and monitor user orders.

METHODOLOGY: WATERFALL MODEL

Reason for Choice

- The **Waterfall Model** was selected because it is **simple, linear, and easy to manage** for a small academic project.
- Each stage follows a clear sequence, making documentation and progress tracking straightforward.
- This approach is suitable when the requirements are well-defined and unlikely to change frequently.

METHODOLOGY: WATERFALL MODEL

Stage	Description and Application to Project
1. Requirement Analysis	All functional and non-functional requirements are gathered and documented (Person 1).
2. System Design	UML diagrams and design patterns are created to define the structure (Person 2).
3. Implementation	The main system and modules are developed based on the design (Person 3).
4. Testing	Unit and integration tests are conducted to ensure correctness (Person 4).
5. Deployment & Documentation	The final version is documented, and a presentation is prepared.

Summary

The **Waterfall methodology** provides a structured and organized framework for this online shopping system project.

It ensures a logical progression from requirements to design, implementation, and testing.

By maintaining a clear relationship between each stage, the project team can deliver a consistent and well-documented final product.

